



COMPETENCY STANDARD

FOR

BOILER OPERATION AND MAINTENANCE

(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Boiler Operation and Maintenance** serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored to the shipbuilding sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**, this document addresses the skills requirements critical to boiler operation and maintenance. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (Total 30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-BOM-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-SBD-BOM-02-G	Carry out workplace interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	10
SICIP-SBD-BOM-03-G	Operate in a team environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	10
Total Hour			30 Hrs.

Sector Specific Competencies (Total 30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-BOM-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-SBD-BOM-02-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	20
Total Hour			30 Hrs.

Occupation Specific Competencies (Total 300 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-BOM-01-O	Interpret Basics of Steam Boiler and its Components	1. Identify different types of steam Boiler 2. Understand boiler classification 3. Comprehend heat transfer principles in Boiler	30
SICIP-SBD-BOM-02-O	Understand Boiler Safety and Regulations	1. Interpret codes and standard relating to boiler operation and maintenance 2. Understand room layout and safety devices 3. Follow emergency shutdown procedure	30
SICIP-SBD-BOM-03-O	Perform Boiler Operation	1. Prepare Boiler for operation 2. Start and operate Boiler 3. Monitor and control Boiler performance 4. Shutdown and secure Boiler	100
SICIP-SBD-BOM-04-O	Perform Water Treatment for Boiler Efficiency	1. Understand importance of boiler treatment 2. Identify methods of water treatment 3. Carry out water treatment	40
SICIP-SBD-BOM-05-O	Perform Routine Maintenance	1. Inspect gaskets, seals, and insulation 2. Test and maintain safety valves 3. Test and maintain electrical components 4. Clean boiler tubes, heat exchangers and combustion chamber	50
SICIP-SBD-BOM-06-O	Diagnose and Rectify Boiler Faults	1. Diagnose and address boiler faults 2. Identify and rectify leaks	50
Total Hours			300 Hrs.

COMPETENCY STANDARD: BOILER OPERATION AND MAINTENANCE

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-BOM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 1.5 Common health issues are recognized. 1.6 Common safety issues are identified and applied.
3. Report hazards and risks	1.1 Hazards and risks are identified. 1.2 Hazards and risks assessment and controls are interpreted 1.3 Hazards and risk assessment are reported
4. Respond to emergencies	1.1 Respond to alarms and warning devices. 1.2 <u>Emergency response plans and procedures</u> are responded to. 1.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective equipment	2.1 Safety glasses

	2.2 Ear plugs 2.3 Gloves 2.4 Apron 2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	1.8 Identifying OHS policies and procedures 1.9 Applying personal health and safety practices 1.10 Reporting hazards and risks 1.11 Responding to emergencies
3. Underpinning Attitudes	1.12 Commitment to occupational health and safety 1.13 Promptness in carrying out activities 1.14 Sincere and honest to duties 1.15 Environmental concerns 1.16 Eagerness to learn 1.17 Tidiness and timeliness 1.18 Respect for rights of peers and seniors in the workplace 1.19 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.20 Workplace (simulated or actual) 1.21 Workplace documents, signs and symbols 1.22 Codes of conduct 1.23 Projector 1.24 Learning manual

	1.25 Tools, equipment and facilities appropriate to the process or activities. 1.26 Materials are relevant to the proposed activity.
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-BOM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	Range (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening

	1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	1.1 Notes 1.2 Arranging a meeting 1.3 Agenda 1.4 Simple reports such as progress and incident reports 1.5 Job sheets 1.6 Operational manuals 1.7 Brochures and promotional material 1.8 Visual and graphic materials 1.9 Standards 1.10 OHS information 1.11 Signs
3. Appropriate sources	1.1 Human Resources (HR) Department 1.2 Managers 1.3 Supervisors 1.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	1.5 Demonstrating workplace communication and etiquette 1.6 Interpreting workplace instructions and symbols 1.7 Demonstrating active participation in workplace meeting 1.8 Applying professional ethics at work
3. Underpinning Attitudes	1.9 Commitment to occupational health and safety 1.10 Promptness in carrying out activities 1.11 Sincere and honest to duties 1.12 Environmental concerns 1.13 Eagerness to learn 1.14 Tidiness and timeliness 1.15 Respect for rights of peers and seniors in the workplace 1.16 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-BOM-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
1. Sharing information	1.1. Agenda 1.2. Minutes

	1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information
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Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	1.1. Identifying own role and responsibilities within team 1.2. Communicating and co-operating with team members 1.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	1.4. Commitment to occupational health and safety 1.5. Promptness in carrying out activities 1.6. Sincere and honest to duties 1.7. Eagerness to learn 1.8. Tidiness and timeliness 1.9. Environmental concerns 1.10. Respect for rights of peers and seniors at workplace 1.11. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 1.12. Workplace (simulated or actual) 1.13. Workplace documents, Projector 1.14. Tools, equipment and facilities appropriate to the process or activities. 1.15. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated

	work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN SHIPBUILDING SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-BOM- 01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in shipbuilding sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Green concepts in the shipbuilding sector</u> are understood. 1.2 <u>Energy-efficient technologies</u> are recognized. 1.3 <u>Use of eco-friendly materials</u> is understood. 1.4 <u>Pollution control measures</u> are interpreted. 1.5 <u>Regulatory compliance</u> with regulations and conventions is identified. 1.6 <u>Energy recovery systems</u> are identified as methods to improve energy efficiency.
2. Minimize resources use in the workplace	2.1 <u>Resource usage in shipbuilding processes</u> is identified. 2.2 Water consumption methods are identified through the installation of efficient water management systems. 2.3 <u>Process improvements</u> are understood. 2.4 Compliance with environmental regulations and industry standards is interpreted.
3. Implement waste management practices	3.1 <u>The types of waste in shipbuilding</u> are identified and described. 3.2 <u>Environmental impacts of improper waste disposal</u> in shipbuilding are understood. 3.3 <u>Principles of waste hierarchy</u> are understood for sustainable waste management. 3.4 <u>Importance of safe storage and containment of waste materials</u> is understood.

Range of Variables

Variable	Range
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	May include but not limited to:
1. Green concepts in the shipbuilding sector	1.1 Eco-friendly Hull Design 1.2 Use of Sustainable and Recyclable Materials 1.3 Waste Reduction and Management Practices 1.4 Emission Control and Air Pollution Reduction 1.5 Ballast Water Treatment and Marine Ecosystem Protection
2. Energy-efficient technologies	2.1 Hybrid and Electric Propulsion Systems 2.2 Advanced Hull Coatings to Reduce Drag 2.3 Waste Heat Recovery Systems 2.4 Air Lubrication Systems 2.5 Energy-efficient LED Lighting and Automation 2.6 Smart Energy Management and Monitoring Systems
3. Use of eco-friendly materials	3.1 Recycled Steel and Aluminum 3.2 Biodegradable Composites and Polymers 3.3 Sustainably Sourced Timber 3.4 Non-toxic Anti-fouling Coatings 3.5 Natural Fiber Reinforced Plastics 3.6 Low VOC (Volatile Organic Compounds) Paints and Sealants
4. Pollution control measures	4.1 Prevention of water 4.2 Prevention of air 4.3 Prevention of noise pollution
5. Regulatory compliance	5.1 International Maritime Organization (IMO) Regulations 5.2 Environmental Protection Agency (EPA) Standards 5.3 Marine Pollution (MARPOL) Convention Compliance 5.4 Occupational Health and Safety (OHS) Regulations 5.5 Ship Recycling and Basel Convention Guidelines 5.6 Local and National Environmental Permitting
6. Energy recovery systems	6.1 Waste Heat Recovery Boilers (WHRB) 6.2 Thermoelectric Energy Conversion Systems 6.3 Exhaust Gas Recirculation (EGR) Systems 6.4 Regenerative Braking for Hybrid Propulsion
7. Resource usage in shipbuilding processes	7.1 Raw Material Consumption (Steel, Aluminum, Composites) 7.2 Water Usage in Fabrication and Painting 7.3 Energy Consumption During Manufacturing 7.4 Use of Recycled and Sustainable Materials
8. Process improvements	8.1 Automation and Robotics in Assembly 8.2 Modular Construction and Prefabrication 8.3 Digital Twin and 3D Modeling Integration 8.4 Just-In-Time (JIT) Inventory Management 8.5 Enhanced Quality Control and Testing Procedures
9. Types of waste in shipbuilding	9.1 Hazardous wastes 9.1.1 Paints 9.1.2 Solvents 9.1.3 Oils

	9.1.4 Asbestos 9.2 Non-hazardous wastes 9.2.1 Metal scraps 9.2.2 Wood and general refuse
10. Environmental impacts of improper waste disposal	10.1 Including soil contamination 10.2 Water pollution, and air quality degradation, are understood.
11. Principles of waste hierarchy	11.1 Reduction 11.2 Reuse 11.3 Recycling 11.4 Recovery 11.5 Disposal
12. Importance of safe storage and containment of waste materials	12.1 Prevent leaks 12.2 Prevent spills 12.3 Environmental contamination

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental Green Concepts and Sustainability Principles in Shipbuilding 1.2 Overview of Energy-Efficient Technologies and Their Applications 1.3 Types and Benefits of Eco-Friendly Materials in Shipbuilding 1.4 Key Pollution Control Measures and Their Implementation 1.5 Regulatory Frameworks, Conventions, and Compliance Requirements 1.6 Energy Recovery Systems and Their Role in Enhancing Efficiency 1.7 Resource and Water Usage Management in Shipbuilding Processes 1.8 Understand of environmental Impacts, and Sustainable Practices
2. Underpinning Skills	2.1 Applying Green Concepts in Shipbuilding Practices 2.2 Evaluating and Selecting Energy-Efficient Technologies 2.3 Identifying and Using Eco-Friendly Materials Effectively 2.4 Implementing Pollution Control Measures Onsite 2.5 Interpreting and Ensuring Regulatory Compliance 2.6 Operating and Maintaining Energy Recovery Systems 2.7 Monitoring and Managing Resource and Water Usage 2.8 Improving Shipbuilding Processes for Environmental Sustainability 2.9 Classifying, Handling, and Safely Storing Shipbuilding Waste
3 Underpinning	3.1 Commitment to occupational health and safety

Attitudes	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: USE HAND AND POWER TOOLS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-BOM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate <u>hand tools</u> are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 <u>Measuring tools</u> are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate <u>power tools</u> are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage.

	4.5. Defective hand and power tools are inspected and repaired or replaced. 4.6. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square 3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification
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	1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET BASICS OF STEAM BOILER AND ITS COMPONENTS	Nominal Duration: 30 Hrs.	Unit Code: SICIP-SBD-BOM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret basics of steam boiler and its components. It specifically includes the task of Identifying different types of steam boiler, understanding boiler classifications and comprehending heat transfer modes in boiler.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify different types of steam Boiler	1.1 Different <u>types of steam boilers</u> are identified 1.2 Functions of steam boilers are explained 1.3 <u>Boiler mountings</u> are identified 1.4 Functions of boiler mounting are explained.
2. Understand boiler classifications	2.1 <u>Boiler classifications</u> are identified 2.2 Each type of boilers is explained
3. Comprehend heat transfer modes in Boiler	3.1 <u>Heat transfer modes</u> in boiler are identified 3.2 Each of the modes is explained

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of steam boilers	1.1 Water tube 1.2 Fire tube
2. Boiler mountings	2.1 Safety valves. 2.2 Main steam stop valve 2.3 Bypass valve 2.4 Feed water valve 2.5 Gauge glass 2.6 Blowdown valve 2.7 Vent valve 2.8 Scum valve 2.9 Sampling valve 2.10 Inspection hole/access doors 2.11 Drain valves 2.12 Soot blower
3. Boiler classifications	3.1 Fuel type 3.1.1 Gas fuel 3.1.2 Liquid fuel 3.1.3 Solid fuel

	3.2 Low pressure 3.3 High pressure 3.4 According to mobility 3.4.1 Stationary 3.1.2 Portable
4. Heat transfer modes	3.1 Conduction 3.2 Convection 3.3 Radiation

Curricular Content Guide

1. Underpinning Knowledge	1.1 Different types of steam Boiler 1.2 Functions of steam boilers 1.3 Functions of boiler mountings 1.4 Understand boiler classifications 1.5 Explain each type of boilers 1.6 Comprehend heat transfer modes in Boiler 1.7 Explained each of the modes
2. Underpinning Skills	2.1 Identifying different types of steam Boiler 2.2 Identifying Boiler mountings 2.3 Identifying boiler classifications 2.4 identifying Heat transfer modes in boiler
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere & honest to duties. 3.4 Eagerness to learn. 3.5 Tidiness & timeliness. 3.6 Environmental concerns. 3.7 Respect for rights of peers and seniors at workplace. 3.8 Communication with peers and seniors at workplace
4 Resource Implications	The following resources must be provided: 4.1 workplace (actual or simulate) 4.2 materials relevant to proposed activities 4.3 tools, equipment and documentation 4.4 specifications or work instructions

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified different types of steam Boiler 1.2 Identified Boiler mountings 1.3 identified boiler classifications 1.4 identified heat transfer modes in boiler
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: UNDERSTAND BOILER SAFETY AND REGULATIONS	Nominal Duration: 30 Hrs.	Unit Code: SICIP-SBD-BOM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand boiler safety and regulations. It specifically includes the task of Interpreting codes and standard relating to boiler operation and maintenance, understanding room layout and safety devices and following emergency shutdown procedure.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret codes and standard relating to boiler operation and maintenance	1.1 Act and policies are identified 1.2 Codes and standard of boiler are interpreted 1.3 Rules and regulation are interpreted
2. Understand room layout and safety devices	2.1 <u>Room layout</u> and <u>safety devices</u> are identified 2.2 Safety precautions for boiler operation are interpreted 2.3 Boiler related fire safety is explained
3. Follow emergency shutdown procedure	3.1 <u>Emergency situations</u> are recognized 3.2 <u>Emergency shutdown procedures</u> are identified 3.3 Case study on boiler accident is demonstrated

Range of Variables

Variable	Range (Includes but not limited to):
1. Room layout	1.1 Safety 1.2 Good ventilation 1.3 Easy access 1.4 Efficient flow of steam, water and fuel
2. Safety devices	2.1 Safety interlocks 2.2 Safety valves 2.3 Gauge glass 2.4 Pressure gauge 2.5 Trip device (high steam, low steam, low water level, high water level) 2.6 Blow down valve 2.7 Flame failure device 2.8 Vent valve 2.9 Temperature control valve 2.10 Safety alarm
3. Emergency situations	3.1 Flame failure 3.2 Low water level

	3.3 High water level 3.4 High temperature 3.5 Unusual noise 3.6 Fuel, steam and water leakage 3.7 Tube failure
4. Emergency shutdown procedures	4.1 Stopping operation by using shutdown button 4.2 Closing fuel cutoff valve 4.3 Operating easing gear of safety valve

Curricular Content Guide

1. Underpinning Knowledge	1.1 Act and policies 1.2 Codes and standard of boiler 1.3 Rules and regulation 1.4 Room layout and safety devices 1.5 Emergency situations 1.6 Emergency shutdown procedure
2. Underpinning Skills	2.1 identifying act and policies 2.2 interpreting codes and standard of boiler 2.3 interpreting rules and regulation 2.4 identifying room layout and safety devices 2.5 recognizing emergency situations 2.6 identifying emergency shutdown procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationary 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified act and policies 1.2 interpreted codes and standard of boiler 1.3 interpreted rules and regulation 1.4 identified room layout and safety devices 1.5 recognized emergency situations
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	1.6 identified emergency shutdown procedures
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM BOILER OPERATION	Nominal Duration: 100 Hrs.	Unit Code: SICIP-SBD-BOM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required perform boiler operation. It specifically includes the tasks of preparing boiler for operation, starting and operate boiler monitoring and control boiler performance and shutting down and securing Boiler.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare Boiler for Operation	1.1 <u>Pre-start checks</u> and equipment readiness are performed 1.2 Boiler is filled with water as per specification 1.3 <u>Firing sequence and ignition procedures</u> are followed.
2. Start and operate Boiler	2.1 Warm-up of boiler water is carried out. 2.2 Load ramping is performed maintaining the steps of ramping. 2.3 Load shedding is done decreasing loads. 2.4 Boiler is operated maintaining running parameters.
3. Monitor and control Boiler performance	3.1 Steam pressure & temperature, water level are checked and controlled. 3.2 Feed water pressure, fuel pressure, fuel temperature, flame colour are checked and controlled. 3.3 Water level of feed tank is checked and maintained. 3.4 All boiler drains & traps are checked and maintained.
4. Shutdown and secure Boiler	4.1 Gradual load decreasing is performed according to manufacturer's guideline 4.2 Burner and fuel system are switched off 4.3 Post purge is performed 4.4 Boiler is secured when pressure stands below one bar 4.5 Vent valve is made open

Range of Variables

Variable	Range (Includes but not limited to):
1. Pre-start checks	1.1 Checking manhole and handhole 1.2 Inspection of fuel systems 1.3 Checking safety devices 1.4 Checking of water level 1.5 Inspecting all instruments and control systems
2. Firing sequence and ignition procedures	2.1 Pre-Purge

	2.2 Pilot Flame Ignition (or Spark Ignition) 2.3 Pilot Flame Proving 2.4 Main Burner Ignition 2.5 Main Flame Proving 2.6 Run / Modulation
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Pre-start checks and equipment readiness 1.2 Firing sequence and ignition procedures 1.3 Warm-up, load ramping, and load shedding 1.4 Boiler operation procedure 1.5 Burner and fuel system
2. Underpinning Skills	2.1 performing pre-start checks and equipment readiness 2.2 following firing sequence and ignition procedures 2.3 performing warm-up, load ramping, and load shedding 2.4 operating boiler maintaining running parameters 2.5 operating shutdown procedure
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationery 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 performed pre-start checks and equipment readiness 1.2 followed firing sequence and ignition procedures 1.3 performed warm-up, load ramping, and load shedding 1.4 operated boiler maintaining running parameters 1.5 operated shutdown procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning

	2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM WATER TREATMENT FOR BOILER EFFICIENCY	Nominal Duration: 40 Hrs.	Unit Code: SICIP-SBD-BOM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform water treatment for boiler efficiency. It specifically includes the tasks of understanding importance of boiler treatment, identifying methods of water treatment and carrying out water treatment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand importance of boiler treatment	1.1 <u>Boiler water treatments</u> are explained. 1.2 <u>Types of boiler water impurities and their effects</u> are recognized.
2. Identify methods of water treatment	2.1 <u>Methods of water treatment</u> are identified. 2.2 Water treatment methods are described.
3. Carry out water treatment	3.1 Testing for boiler water treatment are done. 3.2 Boiler treatments methods are applied. 3.3 Water treatment is performed.

Range of Variables

Variable	Range (Includes but not limited to):
1. Boiler water treatment	1.1 Testing pH 1.2 Importance of maintaining pH 1.3 Testing P-Alkalinity, M-Alkalinity & total Alkalinity 1.4 Testing of Total Dissolved Solid (TDS) 1.5 Testing of chlorides 1.6 Testing of phosphates 1.7 Importance of maintaining phosphate reserve 1.8 Testing of hydrazine and importance maintaining reserve. 1.9 Testing of oxygen 1.10 Testing of hardness
2. Types of boiler water impurities and their effects	2.1 Suspended solids – foaming & carry over. 2.2 Chlorides – scale formation & corrosion. 2.3 Silica – scale formation 2.4 Oil – foaming, carry over, forming acid
3. Methods of water treatment	3.1 De-aeration 3.2 Blowdown 3.3 Softening/Chemical dosing

Curricular Content Guide

1. Underpinning Knowledge	1.1 Boiler water treatments 1.2 Types of boiler water impurities and their effects 1.3 Methods of water treatment
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	1.4 Testing for boiler water treatment
2. Underpinning Skills	2.1 explaining Boiler water treatments 2.2 recognizing types of boiler water impurities and their effects are 2.3 identifying methods of water treatment 2.4 applying boiler treatments methods
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationary 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 explained Boiler water treatments 1.2 recognized types of boiler water impurities and their effects 1.3 identified methods of water treatment 1.4 applied Boiler treatments methods
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ROUTINE MAINTENANCE	Nominal Duration: 50 Hrs.	Unit Code: SICIP-SBD-BOM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Perform routine maintenance. It specifically includes the tasks of inspecting gaskets, seals, and insulation, testing and maintaining safety valves and testing and maintaining electrical components and cleaning boiler tubes, heat exchangers and combustion chamber.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect gaskets, seals, and insulation	1.1 Gaskets, seals & insulations are checked for tightness and leaks. 1.2 Sliding feet is inspected for lubrication and easy movement.
2. Test and maintain safety valves	2.1 Safety valves are tested as per design. 2.2 Maintenance of safety valves are performed as per regulations. 2.3 Easing gear of safety valve is checked for smooth operation. 2.4 Leaks of safety valves is checked.
3. Test and maintain electrical components	3.1 Testing of <u>electrical components</u> are performed. 3.2 Repairing of faulty devices is performed. 3.3 Maintenance of electrical components are carried out as per manual.
4. Clean boiler tubes, heat exchangers and combustion chamber	4.1 Boiler is taken out of service and inspection activities are prepared. 4.2 <u>Safety precautions</u> for inspection are identified. 4.3 <u>Boiler components</u> for inspection are identified. 4.4 <u>Cleaning methods</u> are identified. 4.5 Boiler tubes, heat exchangers and combustion chamber are cleaned.

Range of Variables

Variable	Range (Includes but not limited to):
1. Electrical components	1.1 Testing of tripping devices (Relay, solenoid, sensors, limit switch) 1.1.1 Device for low water level 1.1.2 Device for high steam pressure 1.1.3 Device for flame failure 1.1.4 Device for low fuel pressure 1.1.5 Device for high temperature 1.2 Testing device for alarms –

	1.2.1 Device for high water level 1.2.2 Device for low water level 1.2.3 Device for high steam pressure 1.2.4 Device for low fuel pressure 1.2.5 Device for high temperature.
2. Safety precautions	2.1 Gas freeing 2.2 Proper ventilation 2.3 Recording oxygen level 2.4 Lockout & tag out 2.5 Obtaining permit to enter
3. Boiler components	3.1 Inspecting water side 3.1.1 Inspect for hard deposits 3.1.2 Inspect for corrosion 3.1.3 Inspect for cracks 3.1.4 Inspect for cleanliness 3.2 Inspecting fire side 3.2.1 Inspect for hard deposits 3.2.2 Inspect for refractories damage 3.2.3 Inspect for cracks 3.2.4 Inspect for cleanliness 3.3 Inspecting burners 3.3.1 Inspect burner nozzle 3.3.2 Inspect air register
4. Cleaning methods	4.1 Soot blowing 4.2 Descaling 4.3 Cleaning with air 4.4 Cleaning with chemicals 4.5 Cleaning with waterjet

Curricular Content Guide

1. Underpinning Knowledge	1.1 Gaskets, seals, and insulation 1.2 safety valves 1.3 electrical components 1.4 boiler tubes, heat exchangers and combustion chamber
2. Underpinning Skills	2.1 Inspecting gaskets, seals, and insulation 2.2 Testing and maintaining safety valves 2.3 Testing and maintaining electrical components 2.4 Cleaning boiler tubes, heat exchangers and combustion chamber
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the

	workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationery 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Inspected gaskets, seals, and insulation 1.2 Tested and maintained safety valves 1.3 Tested and maintained electrical components 1.4 Cleaned boiler tubes, heat exchangers and combustion chamber
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DIAGNOSE AND RECTIFY BOILER FAULTS	Nominal Duration: 50 Hrs.	Unit Code: SICIP-SBD-BOM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to diagnose and rectify boiler faults. It specifically includes the tasks of diagnosing and addressing boiler faults and identifying and rectifying leaks.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Diagnose and address boiler faults	1.1 <u>Common boiler faults</u> are understood 1.2 Their <u>causes</u> are identified 1.3 Boiler faults are diagnosed. 1.4 Diagnosed faults are rectified as per manual.
2. Identify and rectify leaks	1.1 <u>Different types of leaks</u> are identified. 1.2 Method to rectify leaks are understood. 1.3 Diagnosed leaks are rectified.

Range of Variables

Variable	Range (Includes but not limited to):
1. Common boiler faults	1.1 Flame failure 1.2 Water level dropping 1.3 Steam pressure dropping 1.4 Incorrect water level in gauge glass 1.5 Foaming
2. Causes	2.1 Dirty/faulty flame eye 2.2 Low fuel temperature 2.3 Incorrect air fuel ratio 2.4 High TDS 2.5 Oil contamination 2.6 Chocked gauge glass 2.7 Blowdown valve leaking 2.8 Low level in feed tank water
3. Different types of leaks	3.1 Water leaks 3.2 Steam leaks 3.3 Fuel leaks

Curricular Content Guide

1. Underpinning Knowledge	1.1 Common boiler faults 1.2 Diagnose faults and rectification 1.3 Different types of leaks 1.4 Method to rectify leaks 1.5 Diagnosed leaks
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2. Underpinning Skills	2.1 Identifying common boiler faults 2.2 Diagnosing faults and rectification 2.3 Identifying different types of leaks 2.4 Understanding method to rectify leaks 2.5 Rectified diagnosed leaks
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified common boiler faults 1.2 Diagnosed faults and rectification 1.3 Identified different types of leaks 1.4 Understood method to rectify leaks 1.5 Rectified diagnosed leaks
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Boiler Operation and Maintenance
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Fire Tube Boiler (500 Kg per hour)	1
2.	Fuel tank (2 ton)	1
3.	Fuel pump with accessories	1
4.	Feed water tank (2 ton)	1
5.	Feed water pump with accessories	1
6.	System piping for boiler exhaust (Standard size)	1
7.	System piping for water (Standard size)	1
8.	System piping for fuel (Standard size)	1
9.	System piping for steam with insulation (Standard size)	1
10.	Multimeter	2
11.	Pipe Wrenches (24")	2
12.	Adjustable Wrenches	2
13.	Spanners set	1
14.	LPG cylinder with Propane Gas (35kg)	10
15.	Boiler Water Testing Kit	2
16.	Burner Nozzles	2 set
17.	Boiler Tube Cleaning Brushes, chemicals, pump	2
18.	Temperature Gun or Thermometer	1
19.	Flashlight or Inspection Lamp	2
20.	Pipe and Flue Brushes (with item no.6)	2
21.	Gaskets and Seals (5 square meter)	5
22.	Ignition Electrodes	2 set
23.	Flame Sensors	2
24.	Tool box for boiler	2 set
25.	Safety valve	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Boiler Operation and Maintenance** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.